Advanced Software Design Techniques

Organisation

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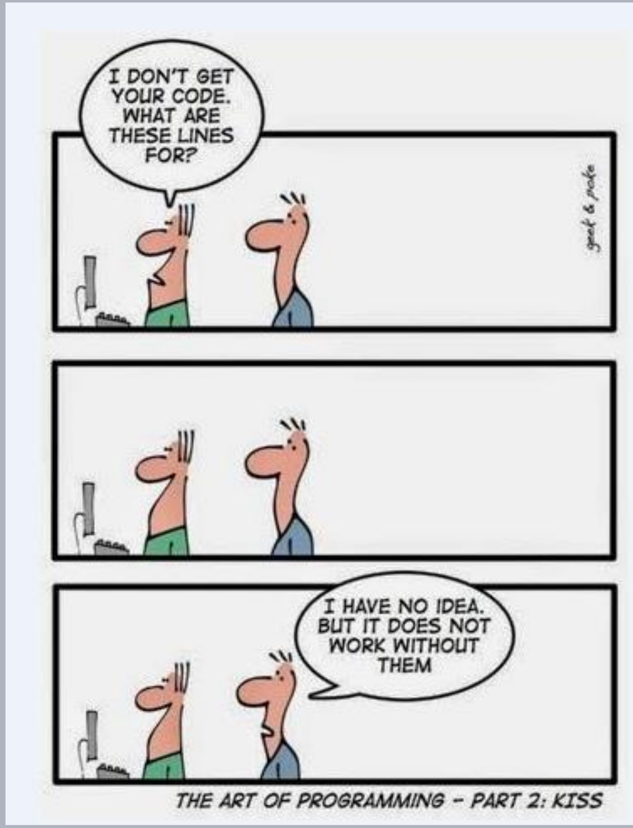


Module Content

Review of fundamental concepts of a widely used object oriented programming language. The course will cover

- advanced data and class structures
- differences and interoperability of C and C++
- polymorphism,
- generic programming,
- introduction to the STL, string and stream library of C++,
- coding standards (MISRA),
- software metrics,
- design patterns,
- refactoring techniques,
- extensions of the C++ standard.

Design aspects like modularity, performance and software re-use will be discussed. Developing software designs using the UML and CASE tools as well as extensive hands-on programming assignments in C/C++ are an integral part of the course.

Learning Outcome

to understand:

- complex design patterns

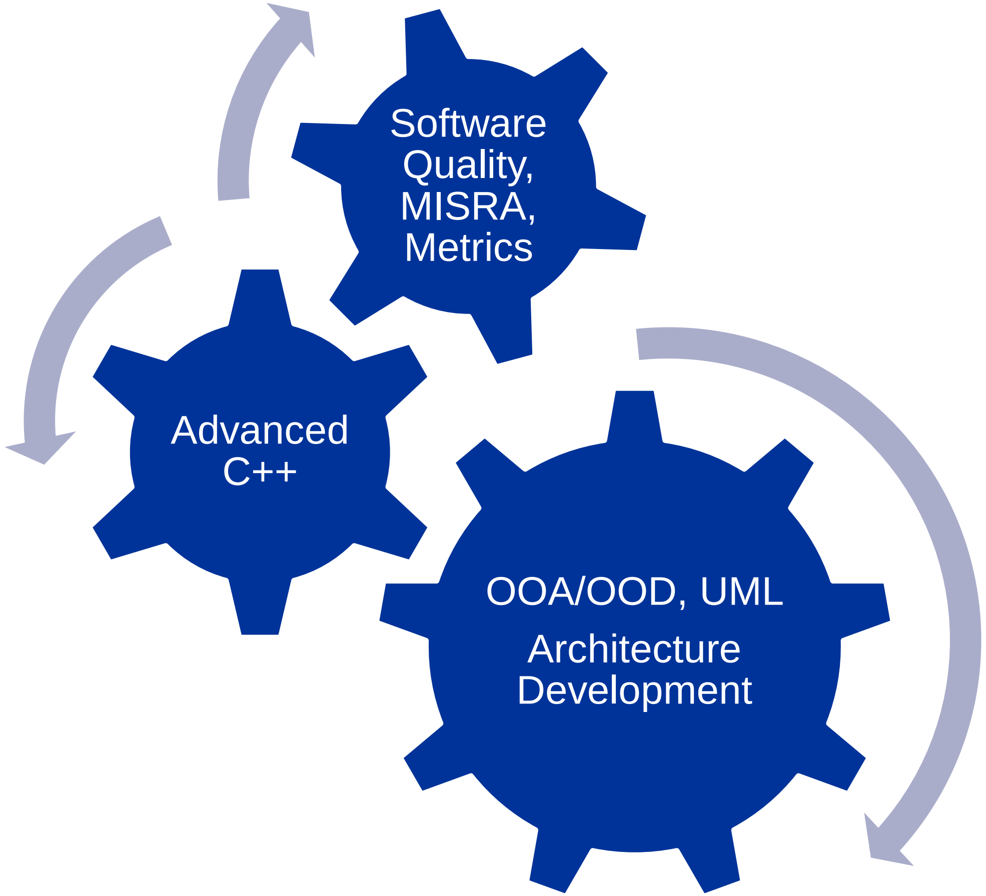
to apply:

- complex design patterns
- assess design quality of complex software
- refactoring methods
- combined C/C++ modules

to transfer:

- the design patterns and concepts to more complex architectures

Advanced Software Design Techniques in a Nutshell



Structure of the Course (draft)

Week	Content	Content
1	Introduction and Motivation	
2	Object Oriented Analysis and Design I	Object Oriented Analysis and Design II
3	Object Oriented Analysis and Design III	
4	Object Oriented Analysis and Design IV	Code Documentation, UML Diagrams
5	Template Programming	
6	Fundamentals of Software Quality and Clean Architecture	Elementary Structural Design Patterns
7	Interface Design (components, classes, methods)	
8	Error Handling / Exceptions	MISRA and Static Code Analysis
9	Metrics	
10	Mixing C and C++	

Lab Assignments

Lab

- 2 mandatory lab@home assignments
  - Design / Architecture
  - Design/Code Evaluation, Refactoring
- Lab has to be passed in order to attend the exam

Tools

- Eclipse (EIT Installation)
- Together
- Doxygen
- Understand C/C++

Exam

- Lab content will be part of the exam
- 90min